Yixuan Huang

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Education

Example University *Ann Arbor, MI*
Ph.D. in Mechanical Engineering *Expected May 2028*

Example Institute of Technology *Boston, MA*
B.S. in Electrical and Computer Engineering *May 2023*

Research Interests

Robotics; autonomous systems; learning-based control; robot perception; human-robot interaction; planning under uncertainty.

Publications

Yixuan Huang, Alex Researcher, and Taylor Advisor *2026*
Learning-Enabled Planning for Robust Robotic Manipulation
Manuscript in preparation

Yixuan Huang and Taylor Advisor *2026*
A Benchmark for Evaluating Perception-Aware Robot Planning
Conference submission under review

Research Experience

Example Robotics Laboratory *Ann Arbor, MI*
Graduate Research Assistant *Sep. 2023 – Present*
Topics: Robotic Manipulation, Planning, Learning-Based Control
Advisor: Prof. Taylor Advisor

- Developed planning and control pipelines for evaluating robotic manipulation under uncertainty.
- Designed simulation experiments to compare task success, execution time, and robustness across methods.

Example Autonomous Systems Group *Boston, MA*
Undergraduate Research Assistant *Jan. 2022 – May 2023*
Topics: Autonomous Navigation, Sensor Fusion, Field Robotics
Advisor: Prof. Casey Mentor

- Implemented data processing tools for analyzing robot trajectories and sensor logs.
- Assisted with field experiments involving mobile robots operating in semi-structured environments.

Selected Projects

Simulation Platform for Robot Rearrangement *2025*
Python, MuJoCo, Motion Planning

- Built a simulation environment for testing tabletop object rearrangement strategies.
- Evaluated methods using task completion time, collision rate, and number of intermediate placements.

Visual Perception Pipeline for Manipulation *2024*
Computer Vision, Depth Sensing, 3D Reconstruction

- Implemented a perception pipeline for estimating object geometry from RGB-D observations.
- Integrated reconstructed scene information into downstream robotic grasp and placement planning.

Learning-Based Control Benchmark *2024*
PyTorch, Reinforcement Learning, Control

- Compared model-free control methods on simulated continuous-control tasks.
- Analyzed policy performance across random seeds, reward designs, and environment perturbations.

Awards & Honors

Graduate Research Fellowship *2025*
Example University
Outstanding Teaching Assistant Award *2024*
Department of Mechanical Engineering, Example University
Dean's List *2020 – 2023*
Example Institute of Technology

Leadership & Service

Graduate Student Mentor *2024 – Present*
Example Robotics Laboratory
Reviewer *2025*
Example Student Research Symposium
Workshop Organizer *2024*
Robotics Reading Group, Example University

Technical Skills

Programming: Python, C/C++, MATLAB, JavaScript
Robotics: ROS, MuJoCo, motion planning, manipulation, perception
Machine Learning: PyTorch, TensorFlow, scikit-learn, reinforcement learning
Tools: Linux, Git, LaTeX, Docker, Jupyter Notebook